A SMALL MOLECULE.
A BIGGER IMPACT.

# CAFFEINE

## The Molecule Behind Alertness

CHEMISTRY BREWS BRIGHTER DAYS.



*Good Chemistry Fuels Great Days*

*"Same molecule. More focus. More possibilities."*

## 1 COMPOUND IDENTITY

**Common name:**
Caffeine

**IUPAC name:**
1,3,7-trimethylpurine-2,6-dione

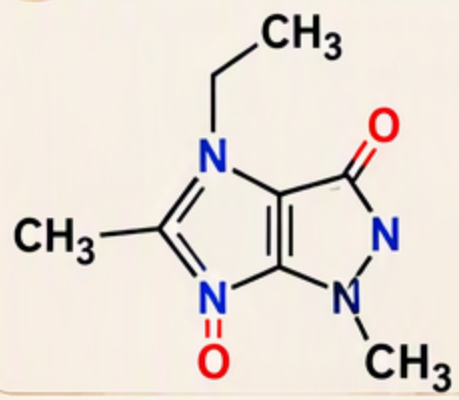
**Molecular formula:**
$C_8H_{10}N_4O_2$

**Molar mass:**
194.19 g/mol

Natural alkaloid

Found in coffee, tea, cacao, and other plant sources

## 2 STRUCTURE




**Key structural features:**

- Nitrogen-containing heterocyclic ring (purine)
- Two carbonyl (C=O) groups
- Three methyl ($-CH_3$) groups
- Classified as a methylxanthine derivative

## 3 IMPORTANT PROPERTIES

**Appearance:** White crystalline solid

**Odor:** Odorless

**Taste:** Bitter

**Solubility:** Soluble in water (≈ 21.6 g/L at 25 °C)

**Melting point:** 238 °C (decomposes)

## 4 WHAT IS AMAZING ABOUT IT?



Caffeine promotes alertness by blocking adenosine receptors in the brain. Adenosine is a chemical that builds up during the day and makes you feel sleepy. By competing with adenosine, caffeine reduces the sensation of tiredness, helping you feel more awake and focused.

*"It doesn't give you energy — it helps you use your energy better."*

## 5 USES AND VALUE



Coffee and tea products



Energy drinks



Pharmaceutical formulations (e.g., pain relievers, alertness aids)



Cacao and chocolate products



Research and laboratory applications

A widely consumed and economically important compound in the food, beverage, and pharmaceutical industries.

## 6 RISKS AND SAFETY



- Excessive intake may cause restlessness, anxiety, and sleep disturbances.
- May increase heart rate and blood pressure in some individuals.
- Not recommended for certain medical conditions (e.g., heart disorders, anxiety disorders) and during pregnancy (high intake).



Moderate consumption is generally considered safe for healthy adults (up to 400 mg per day).

## 7 CHEMICAL ENGINEERING CONNECTION



Caffeine is extracted from coffee, tea, and other sources using solid-liquid extraction, solvent extraction, and adsorption. Chemical engineers design and optimize separation and purification processes such as:

- Extraction (e.g., using water or supercritical $CO_2$)
- Filtration and crystallization
- Adsorption (e.g., activated carbon)
- Solvent recovery and recycling

These operations are essential for producing high-purity caffeine for food, pharmaceutical, and consumer product industries.

## 8 REFERENCES


**PubChem.** (n.d.). Caffeine (CID 2519). National Center for Biotechnology Information. https://pubchem.ncbi.nlm.nih.gov/compound/Caffeine

**National Institute of Standards and Technology.** (n.d.). Caffeine. NIST Chemistry WebBook. https://webbook.nist.gov/cgi/cbook.cgi?Name=caffeine


Explore more on PubChem:

SCAN ME

*From bean to breakthrough.*

*Chemistry today. A brighter tomorrow.*

ORGANIC MOLECULES REAL-WORLD IMPACT